

# Jake Brawer, Ph.D.

[website](#) / [github](#) / [email](#) / [linkedin](#)

## RESEARCH OVERVIEW

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**Summary** I build **interactive AI systems** that combine perception, reasoning, memory, and learning to support robust human-centered interaction. My work spans **robotics**, **conversational AI**, and **production ML systems**, with emphasis on realtime decision-making, structured knowledge, personalization, and deployable architectures.

## PROFESSIONAL EXPERIENCE

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| <b>Founding Engineer</b> | <i>Stealth startup (voice AI / digital health)</i><br>Sole architect and founding engineer for an early-stage phone-based conversational AI platform for older adults.<br>Built the end-to-end system across telephony, realtime speech, LLM orchestration, long-term memory, FastAPI services, PostgreSQL, and AWS infrastructure.<br>Implemented platform safeguards for reliability and HIPAA-oriented handling of sensitive user data.       | Nov 2025–Current |
| <b>Postdoc</b>           | <i>University of Colorado, Boulder</i><br>Advisors: Alessandro Roncone and Bradley Hayes<br>Lead research on human–robot teaming, assistive robotics, and AI decision support. Develop models for inferring human proficiency, trust, and intent from gaze and behavior in collaborative environments.<br>Build semantic robot perception systems for robust localization in cluttered indoor workspaces using low-cost visual-inertial sensing. | 2023–2025        |
| <b>Lecturer</b>          | <i>University of Colorado, Boulder</i><br>Sole lecturer for Advanced Robotics (CSCI 4302/5302), a graduate-level course covering motion planning, SLAM, optimization, and reinforcement learning.                                                                                                                                                                                                                                                | Spring 2024      |

## EDUCATION

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| <b>Yale</b>   | <b>Ph.D. in Computer Science</b><br>Advisor: Brian Scassellati | 2016–2023 |
| <b>Vassar</b> | <b>B.A. in Cognitive Science; Computer Science minor</b>       | 2012–2016 |

## SKILLS

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| <b>Languages</b> | Python, C/C++, C#, SQL, Git, Emacs, Docker, Linux, $\text{\LaTeX}$                                                                                  |
| <b>AI / ML</b>   | PyTorch, scikit-learn, SciPy, NumPy, Pandas, LangChain, LLM orchestration, RAG / vector retrieval, PGVector, matplotlib, NLTK, Gurobi, Prolog, PDDL |

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| <b>Backend</b>     | FastAPI, PostgreSQL, AWS, async Python, REST APIs, system design, deployment, load testing, observability, HIPAA-oriented system design, PHI-safe logging |
| <b>Voice AI</b>    | Pipecat, Deepgram, AWS Bedrock, Telnyx, Cartesia                                                                                                          |
| <b>Robotics/CV</b> | ROS, Gazebo, RViz, OpenCV, Open3D, Baxter, Sawyer, UR5, Kinect, RealSense, RGB-D cameras                                                                  |
| <b>Soft Skills</b> | Clear technical communication, Collaborative, Fast learner, Leadership                                                                                    |

## SELECTED PUBLICATIONS

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Shivendra Agrawal, **Jake Brawer**, Ashutosh Naik, Alessandro Roncone, and Bradley Hayes (2026). “**ShelfAware: Real-Time Visual-Inertial Semantic Localization in Quasi-Static Environments with Low-Cost Sensors**”. In: *IEEE Robotics and Automation Letters*. Accepted.

Nikhil Hulle, Stéphane Aroca-Ouellette, Anthony J Ries, **Jake Brawer**, Katharina von der Wense, and Alessandro Roncone (2024). “**Eyes on the Game: Deciphering Implicit Human Signals to Infer Human Proficiency, Trust, and Intent**”. In: *33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN 2024)*. Accepted for publication, proceedings pending.

**Jake Brawer**, Debasmita Ghose, Kate Candon, Meiyang Qin, Alessandro Roncone, Marynel Vazquez, and Brian Scassellati (2023). “**Interactive Policy Shaping for Human-Robot Collaboration with Transparent Matrix Overlays**”. In: *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction*. IEEE.

**Jake Brawer**, Meiyang Qin, and Brian Scassellati (2020). “**A causal approach to tool affordance learning**”. In: *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, pp. 8394–8399.

## AWARDS AND SERVICE

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| <b>2025</b>      | <b>Outstanding Reviewer Award</b> , Robotics: Science and Systems ( <b>RSS</b> )                                                                                                                                                 |
| <b>2023-2025</b> | <b>Army Educational Outreach Program (AEOP) Fellowship</b> , University of Colorado Boulder                                                                                                                                      |
| <b>2024</b>      | Area Chair, International Conference on Human–Agent Interaction ( <b>HAI</b> )                                                                                                                                                   |
| <b>2023</b>      | <b>Best paper (technical track)</b> , ACM/IEEE International Conference on Human-Robot Interaction ( <b>HRI</b> ) for “Interactive Policy Shaping for Human-Robot Collaboration”                                                 |
| <b>Workshops</b> | Co-organized the Human-Machine Teaming Paradigm Workshop (2023) and Bridging the Gap: An NSF Workshop on Conversational Agents and Human-Robot Interaction (2018).                                                               |
| <b>Service</b>   | Reviewer for <b>RSS</b> , <b>HRI</b> , <b>IROS</b> , <b>ICRA</b> , <b>Humanoids</b> , <b>IJRR</b> , <b>T-RO</b> , and <b>THRI</b> ; mentored undergraduate and graduate researchers on robotics, HRI, and assistive AI projects. |